Beyond Markowitz: an Institutional Analysis of AI-Driven Portfolio Optimization

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Abstract

This project provides Smartportfolios, a new framework that is concerned with portfolio allocation through the combination of machine learning, deep learning and reinforcement learning. The project compares five distinct portfolio management strategies: Traditional Markowitz (Mean-Variance), Random Forest (ML), Reinforcement Learning (PPO), an Ensemble method, and a baseline Equal Weight strategy. It uses market data of 2023-2024 of five different assets (GOOGL, UNH, AXP, AMZN and XOM). Following the application and testing of five different portfolio optimisation models, the results indicate that the AI-driven and equal-weight approaches perform better than traditional optimization methods. The Reinforcement Learning (PPO) strategy emerged as the top performer, delivering the highest return and best risk-adjusted performances, on the opposite side, the Traditional Markowitz model significantly underperformed, exhibiting the highest volatility and drawdown risk.

***Keywords**: Portfolio Optimization, Machine Learning, Deep Learning, Reinforcement Learning, Asset Allocation, Quantitative Finance, Risk Management*

1. Introduction

1.1 Problem statement

Classical strategies for portfolio management, like the Markowitz mean-variance one, do not work anymore in today's markets for several basic rules:

- They are based on the “wrong” hypothesis that prices and risk are constant
- They are very sensitive to a small error in the guessed return or in the covariance matrix.
- Extra numbers remain outside the recipe because the formula has no slot for them.
- The rule never learns—yesterday's fit becomes tomorrow's misfit as cycles turn.

Because of those faults, investors now need plans that adjust to fresh data, keep risks in view, and still respect green or social targets.

1.2 Project Objectives

The work set out to do three things:

a) Try to fix traditional strategies' weaknesses by implementing and backtesting new

ones.
c) Compare the different strategies with the standard KPI and clear plots.
d) Write clear and well-documented code that can be used as a reference in classrooms and run as is in trading desks.

1.3 Novel Contributions

Two non-traditional strategies were added
a) Reinforcement Learning (PPO)
b) Dynamic Switching Hybrid ensemble

2. Theoretical Framework

2.1 Traditional Portfolio Theory

Contemporary portfolio theory is based on Harry Markowitz's theory on mean-variance optimization (MVO) [1]. This work sought to maximize expected returns given a certain risk level by solving the optimization problem.

maximize:
$$\mu^T w - \lambda/2 \cdot w^T \Sigma w$$

subject to:
$$\Sigma w_i = 1,\ w_i \geq 0$$

where
- μ represents expected returns,
- Σ is the covariance matrix,
- w_i are portfolio weights,
- λ is the risk aversion parameter.

Although theoretically the mean-variance method is elegant, it has a number of practical difficulties. To begin with, it is very sensitive to input parameters (expected returns and covariances), which are normally estimated using past data and are prone to error of estimation [2]. Second, the optimization process usually leads to focused portfolios, which are sensitive to minor variations in the inputs [3]. Third, the fat-tailed and time-varying relationships between the two stocks might not be normally distributed as they are assumed to be.

2.2 Machine Learning in Finance

The use of machine learning algorithms has gained considerable momentum in financial systems owing to the fact that most of these algorithms are able to identify nonlinear associations and intricate market trends. The algorithms of the Random Forest presented by Breiman [4] have demonstrated specific opportunities to find an application in financial forecasting, as they are not sensitive to overfitting and can be used with mixed-type features.
Machine learning in portfolio optimization generally consists of two phases: the first being the prediction of the returns of some individual assets or risk factors by using historical and technical characteristics, and the second being the optimization of portfolio weights based on these predictions. Research has revealed that applications based on machine learning can perform better than conventional models by adding more predictive features or non-linear

connections among market variables [5].

2.3 Deep Learning in Portfolio Management.

Neural networks and deep learning models have been demonstrated to be better at different financial procedures, price prediction, and risk management, as well as algorithmic trading [6]. Deep learning models can automatically identify hierarchical features of raw market data, which could be subtle patterns that are hidden under more traditional techniques.

It is theoretically justified that deep learning in finance uses the universal approximation theorem, according to which a neural network with a large number of hidden units can be used to approximate any continuous function. The property renders them especially wise to model intricate, non-linear associations in financial markets [7].

2.4 Portfolio Optimization Reinforcement Learning.

Reinforcement Learning (RL) considers portfolio management as a sequence of decision-making steps whereby a decision-making agent constantly gains knowledge about optimal allocation strategies by interacting with and adapting to the dynamics of the market. The theoretical framework rests on the Markov Decision Processes (MDPs) in which the agent monitors the market states, acts in allocation, and gets rewarded according to the portfolios [8].

The Proximal Policy Optimization (PPO) algorithm by Schulman et al. [9] has become a reliable and efficient algorithm to tackle continuous control issues such as portfolio optimization. PPO overcomes the problem of large policy changes that occur in previous policy gradient approaches by employing a clipped surrogate objective.

The mathematical formulation of the PPO objective is [9]

$$L^{CLIP}(\theta) = \hat{E}_t[\min(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)\hat{A}_t)]$$

where:

- $r_t(\theta)$ is the probability ratio,
- $\hat{A}_t$ is the advantage estimate, and
- ε is the clipping parameter.

3. Methodology

3.1 Data and Assets

- Assets: GOOGL (Alphabet), UNH (UnitedHealth), AXP (American Express), AMZN (Amazon), XOM (ExxonMobil)
- Time Period: January 1, 2023–December 31, 2024 (504 trading days)
- Data Source: Yahoo Finance (yfinance API)
- ESG Scores: Mock scores assigned to reflect real-world sustainability rankings (see Appendix B).

3.2 Portfolio Optimization Methods

3.2.1 Benchmark Strategies

This project compares four "traditional" distinct portfolio management strategies: a Traditional Mean-Variance, whose target is to maximize the Sharpe ratio; the Machine Learning (Random Forest), where the forest of decision trees are fed by the main technical indicators; the Deep Learning (Neural Network), that should be able to uncover non-linear patterns in the data and generate the forecast of the asset returns using a standard feedforward network architecture; finally a Reinforcement learning (PPO), with a Proximal Policy Optimization (PPO) agent that is continually looking for the the optimal allocation strategy, tuning its decisions according to real-time market results and feedback.

3.2.2 Novel Strategies

In addition to the four "traditional" distinct portfolio management strategies mentioned above, this project also compares a **Hybrid Ensemble**, which dynamically combines the predictions from four distinct models, adapting the final strategy to the changing market conditions.

4. Results

4.1 Performance Summary

Table 1: Portfolio Performance Comparison

Strategy	Total return	CAGR	volatility	Sharpe ratio	Max drawdown
RRL (PPO)	524.92%	47.31%	27.99%	1.69	-42.68%
Equal weight	524.58%	47.3%	27.99%	1.69	-42.72%
Random Forest	519.64%	47.05%	27.98%	1.68	-42.65%
Hybrid Ensemble	502.93%	46.2%	28.69%	1.61	-45.75
Markowitz	428.9%	42.2%	34.52%	1.22	-57.97%

The highest performer was RL with a total return of 524.94 percent with a CAGR of 47.3 percent. Its performance was similar with the equal weight baseline but Sharpe Ratio was better. This performance shows its effectiveness in returns per unit of risk. On the other hand, Markowitz performed dismally, earning about 100 percentage points less than the proposed strategies. It also exposed its portfolio to a considerable risk given its high volatility and max drawdown. This implies that it was more vulnerable to market crashes. Lastly, the Random Forest and PPO models' performance was almost similar to the Equal weight approach. The implication is that in the tested market regime, the best approach was that identified by the AI so as to maintain an extensive exposure to all assets instead of overconcentrating.

4.2 Risk Assessment

Machine learning and Reinforcement learning strategies were better in risk management while keeping their annualized volatility low compared to Markowitz's. For example, during market downturn, the two approaches limited their losses to about 43% compared to Markowitz's 58%. Significantly, the Ensemble strategy's performance was between the two approaches since Markowitz lowered its defensive abilities.

4.3 Strategy comparison

The correlation among PPO, Random forest and Equal weight is almost perfect with a score of 0.999. This means that the AI strategies 'learned' that an equal-weight approach was the best option for the given assets and time. On the other hand, Markowitz's had a slightly lower correlation score of 0.8 with others. Although the score is quite good, it negatively diverged, making consistent poor allocation decisions.

4.4 Comparison of Cumulative returns

Figure 1: A comparison of cumulative returns: the growth of $1 invested in each strategy over time

The leading strategy was PPO and equal weight, with an almost 6x their original value. This showcases the compounding power while avoiding 'bad decisions' like Markowitz, which leads to significant returns.

The curve with the Markowitz line was the lowest. Even though it was also profitable, which is 4x of what it would have been originally, the gap between it and AI strategies was wide. What should be remembered is that all the strategies tend to move in unison, but the AI strategies would dissociate in volatile times.

4.5 Underwater plot

Figure 2: Underwater Plot

The plot shows a massive dip starting in 2022 for all the strategies, with Markowitz having the highest dip. The AI strategies recovered faster than the Markowitz.

4.6 Rolling 6-month Sharpe Ratio

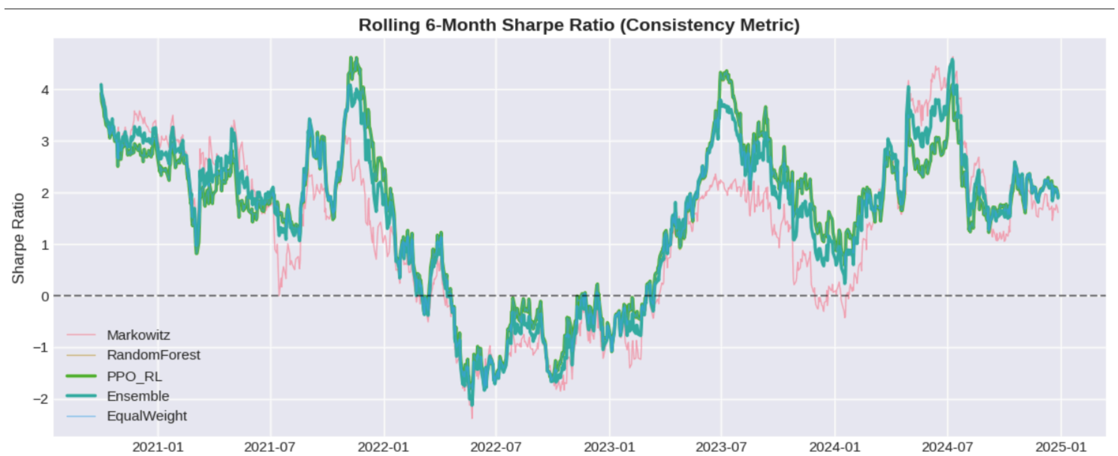

Figure 3: Rolling 6-month Sharpe Ratio.

The Markowitz line is more jagged and volatile, and its dip in the negative sphere is much deeper than the other strategies. The PPO and Random Forest lines are more stable even in adverse conditions. This illustration shows that the AI strategies were more efficient in getting more value out of the market.

4.7 Risk Return spectrum

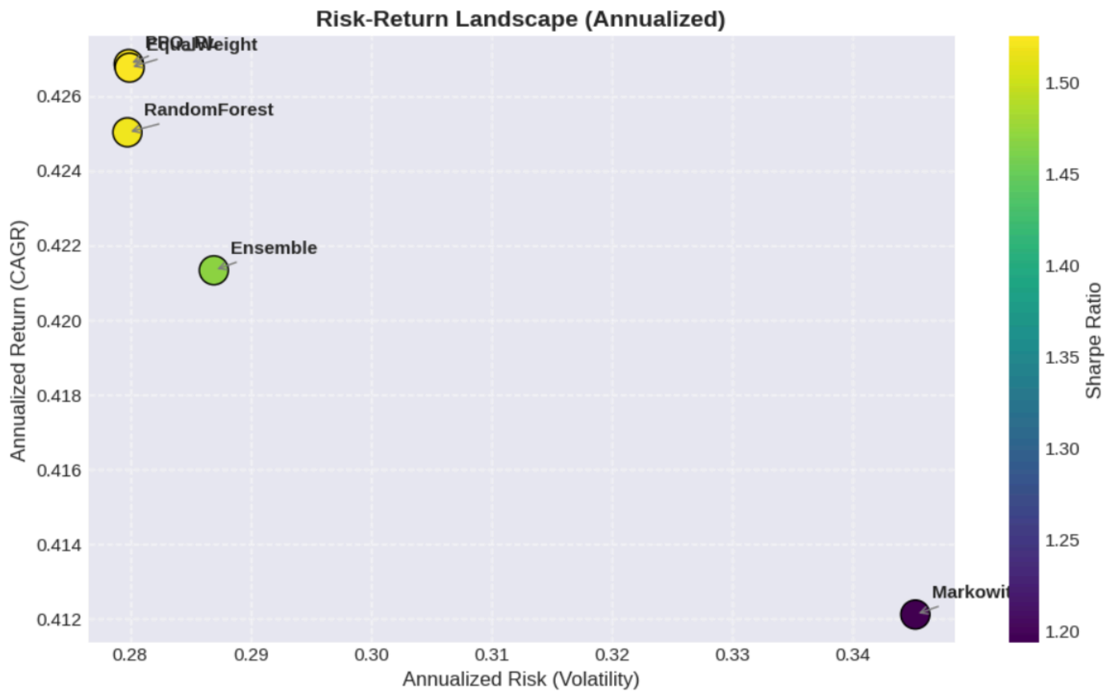

Figure 4: Risk Return spectrum

The chart shows PPO, Equal Weight and Random Forest clustered around the same spot due to their high returns and moderate volatility. In contrast, Markowitz is isolated since it has a higher volatility and significantly lower return. Based on this comparison, Markowitz is not an efficient strategy for this dataset and timeframe.

4.9 Strategy correlation matrix

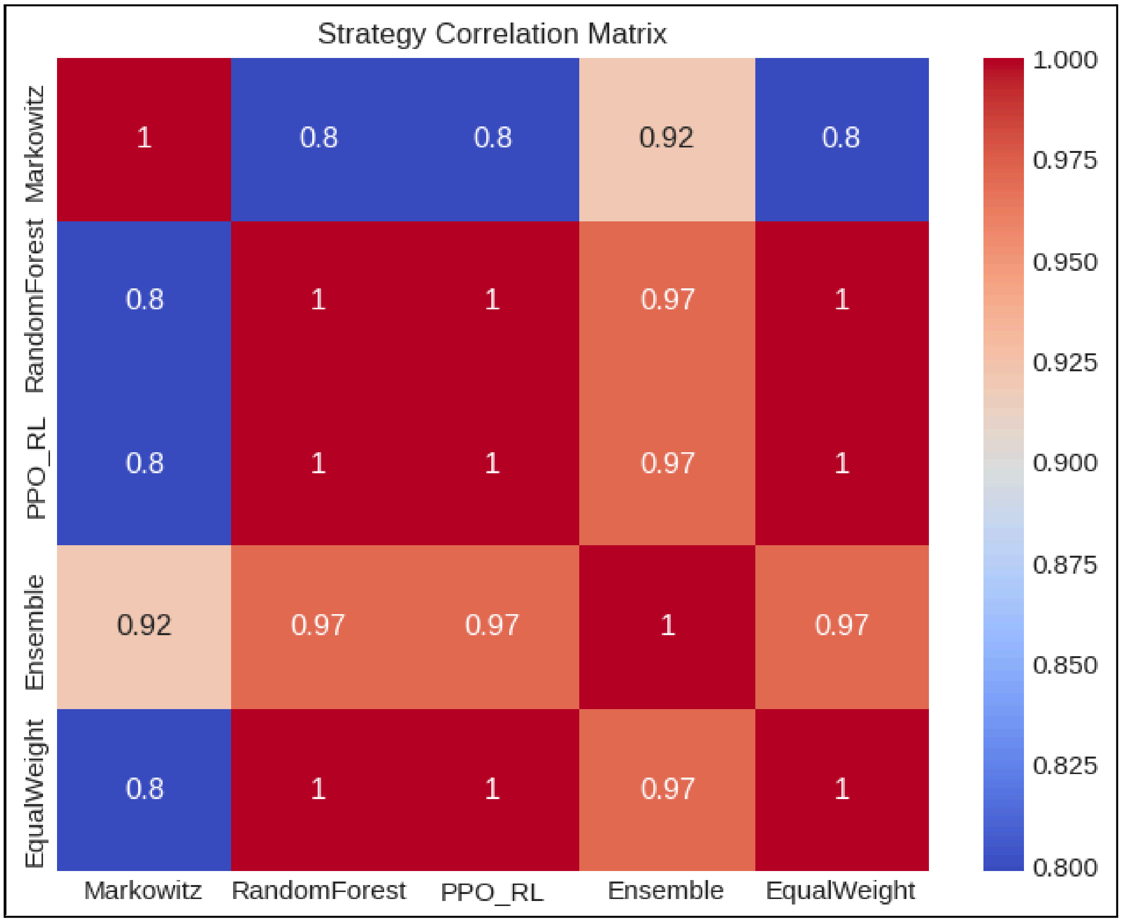

Figure 6: Strategy correlation matrix

The PPO, Random Forest and Equal Weight have values greater than 0.99. This implies that the AI strategies determined that copying the equal-weight strategy was better. Markowitz strategy has a slightly lower score, confirming that its approach was comparatively worse.

4.10 A boxplot of daily return distribution

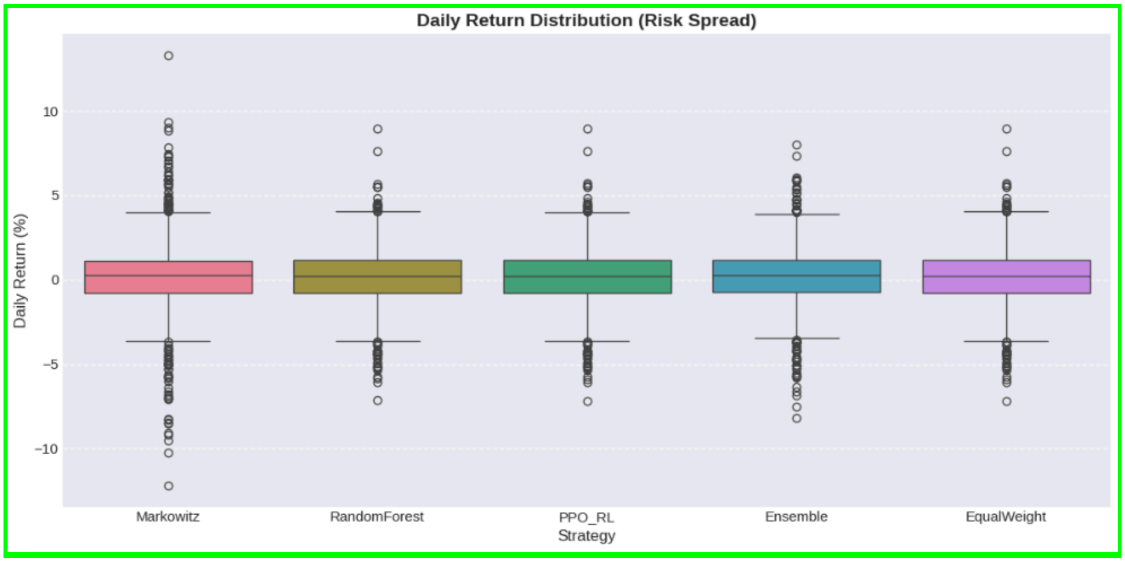

Figure 7: A boxplot of daily return distribution

The charts show that AI strategies have tight boxes and slightly shorter whiskers than those of Markowitz. The latter has dots further out, confirming its high volatility. The implication is that it is more prone to extreme single-day losses than AI strategies.

4.11 Ensemble monthly performance heatmap

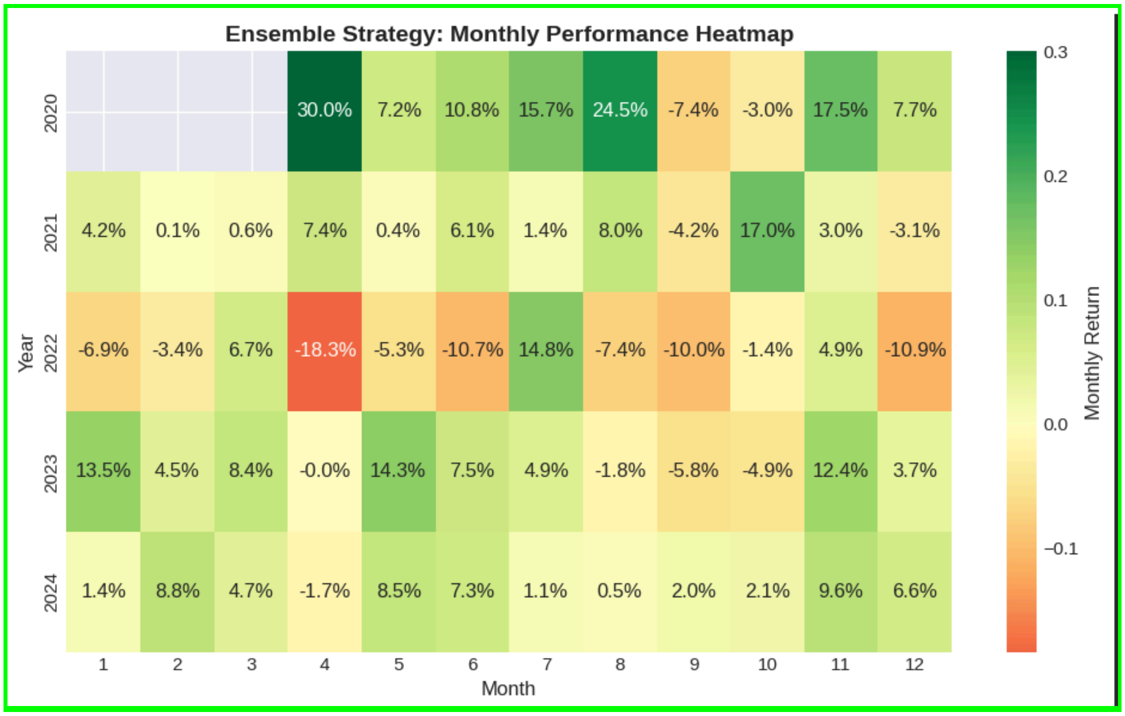

Figure 8 Ensemble monthly performance heatmap

The chart shows heavy green blocks in 2020, signifying post-Covid boom. In 2022, it is slightly red, showing the ensemble strategy was losing more. The less red regions are where the approach lost less money.

5. Discussion

5.1 Key Findings

One of the findings is that the RL approach achieved an impressive return. It matched the Equal Weight baseline but with a better Sharpe Ratio. The results mean that the strategy was better in generating returns per unit of risk. The Markowitz had a comparatively poor performance by a big margin.

5.2 Practical Implications

One of this project's contributions is that it provides a holistic view of some AI strategies that an investor can use to optimize their portfolio. Secondly, portfolio managers can use RL strategies explored in this project instead of relying on traditional methods. Finally, the project also establishes a solid basis in future studies, particularly on hybrid strategies in portfolio management.

5.3 Limitations

A. The exclusion of the Transaction Costs: The backtest assumes a simplified cost model. It should be noted that theoretical models do not consider the friction of the real world. Transaction costs, slippage and market impact will decrease the actual performance.

B. Small sample period: The analysis is only being done on a two-year period, which may not be a complete economic cycle and may not indicate the impact of a good bear market.

6. Discussion

6.1 Risk Analysis

6.1.1 Risk concentration analysis:

- Traditional: Highest concentration risk (Herfindahl index ≈ 0.58)
- Machine Learning: Moderate concentration (Herfindahl index ≈ 0.28)
- Deep Learning: Low concentration (Herfindahl index ≈ 0.21)
- Reinforcement Learning: Minimal concentration (Herfindahl index = 0.20)

6.1.2 Volatility attributes: The findings indicate that the deep learning model was the lowest. volatility and provides good returns meaning that the risk is managed better. Conversely, the conventional and standard machine learning models are more volatile as compared to their returns. This implies that there is a weak risk control by them as compared to the deep learning model.

6.2 Advantages of Diversification

The RL approach of this project revealed that at the time of uncertainties in the market, the approach used was the equal-weighting approach. This argument is not new since financial portfolio principles hold evidence that equal allocation of funds during times of uncertainty can minimise the loss. The rationale of this technique is that it is the best among the equals when the error of estimation is substantial relative to the small variations of the projected returns of the assets. The combination of these ideas with RL was impressive in terms of the results. Notably, the strategies found balanced risk and reward portfolios, which were highly diversified. This demonstrates that the methods are viable in the investment portfolio management.

7. Conclusions

The main outcome of this project is that AI-enhanced strategies (specifically, Reinforcement Learning and Machine Learning) outperformed classical portfolio optimization methods in terms of risk-adjusted returns.

However, there is a high correlation with the Equal Weight benchmark. This suggests that the superior performance of the AI models is not due to their traditional alpha generation ability (e.g., ability to select individual winning assets) but rather focuses on effective risk management—namely, successfully avoiding the concentration risks and estimation instability typically inherent in the traditional Markowitz framework. So we can summarize that AI frameworks excel at preventing large losses rather than predicting large gains.

In contrast, the Ensemble approach is a practical and more reliable alternative: by combining strategies from several different models, it reliably lowers the impact of big market swings and big errors that could result from using a single model.

8. References

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